PROJECT:HIGHLY AVAILABLE WEB APPLICATION WITH AUTO SCALING & MONITORING

Launch an EC2 instance and install Apache or Nginx web server.

deploy a Real time Web application.

Create an AMI from the configured instance.

Create an Application Load Balancer and configure a target group.

Create a Launch Template using the AMI.

Create an Auto Scaling Group with minimum 2 and maximum 4 instances.

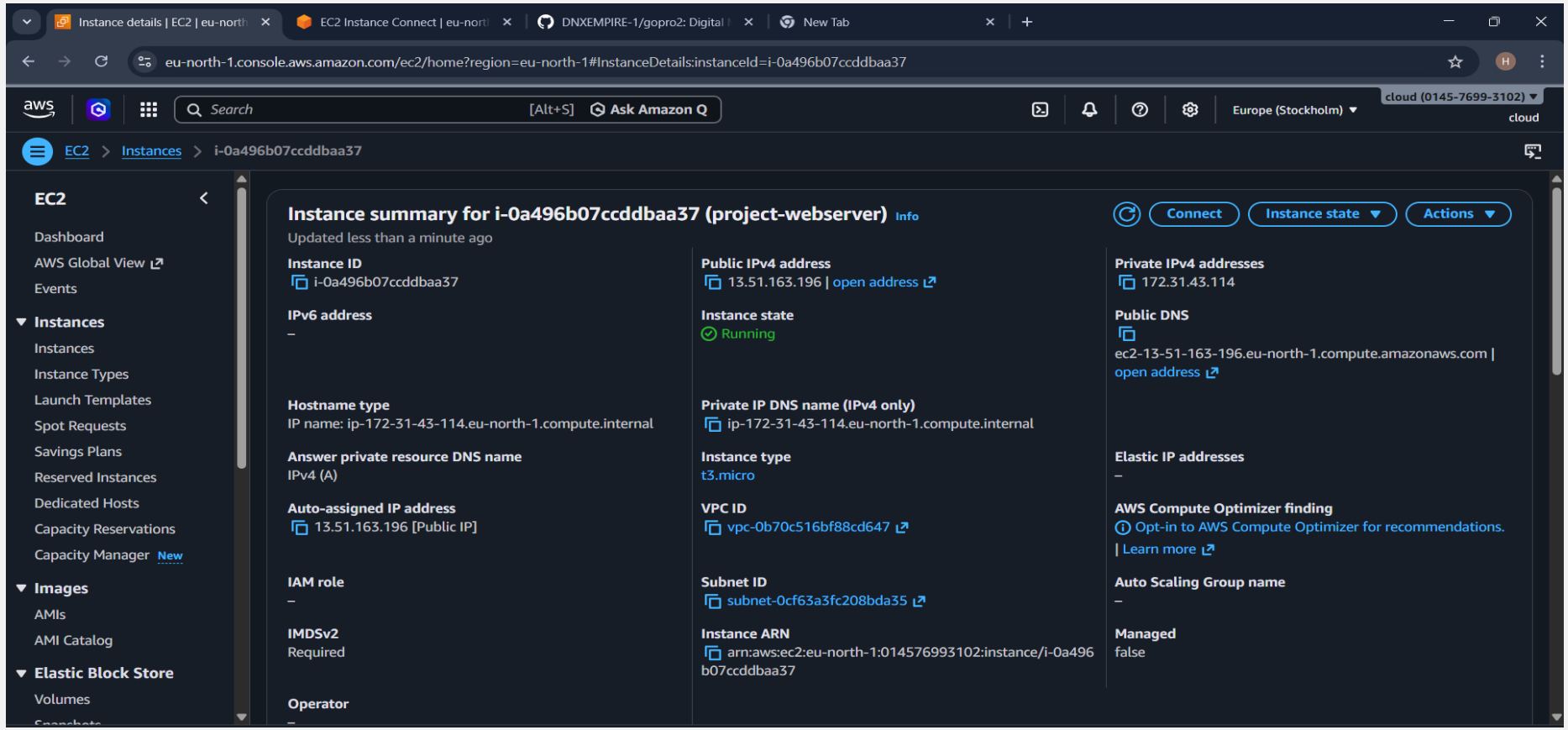
Attach the Load Balancer to the Auto Scaling Group.

Configure scaling policies based on CPU utilization.

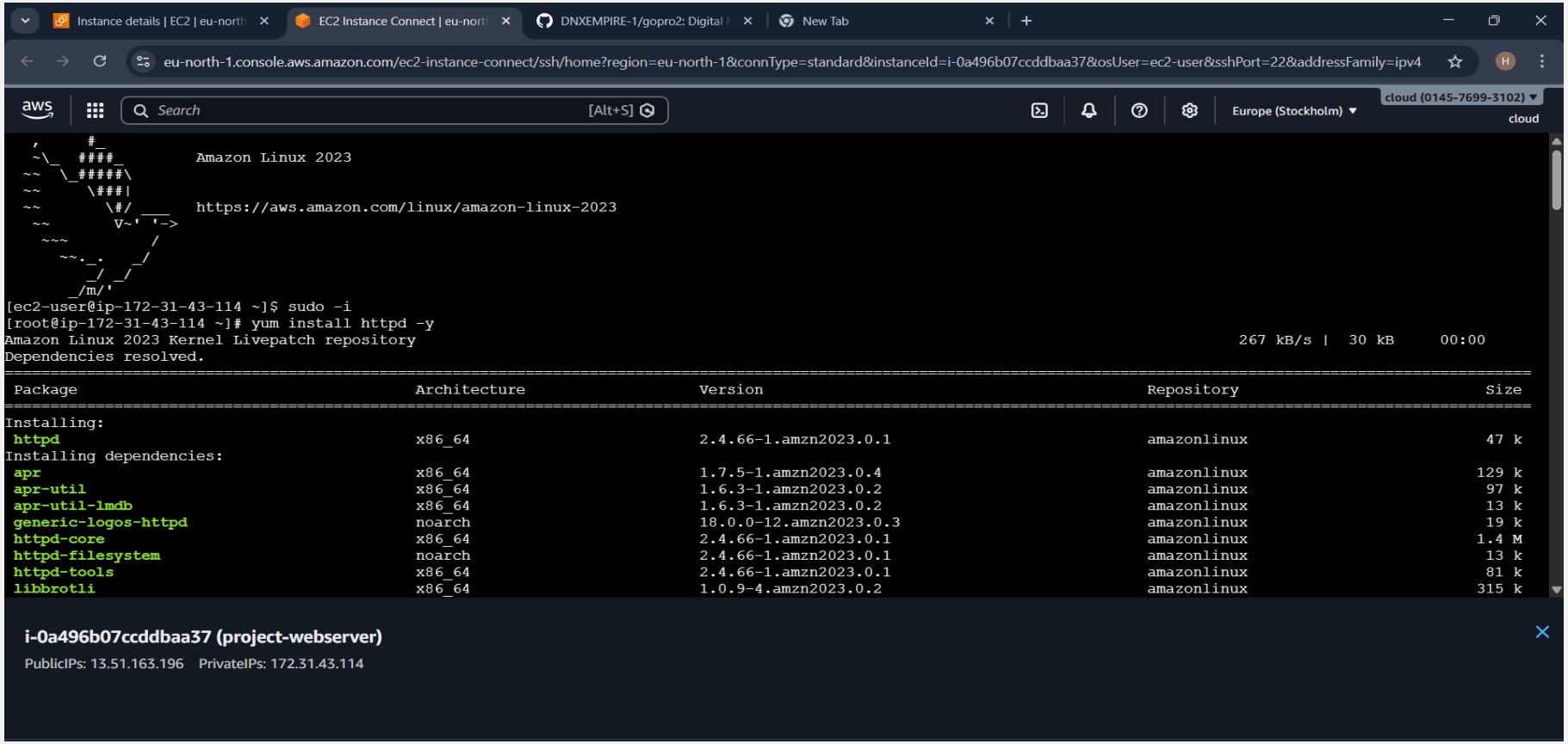
Create CloudWatch alarms to monitor CPU usage.

Create an SNS topic and subscribe using email to receive alerts

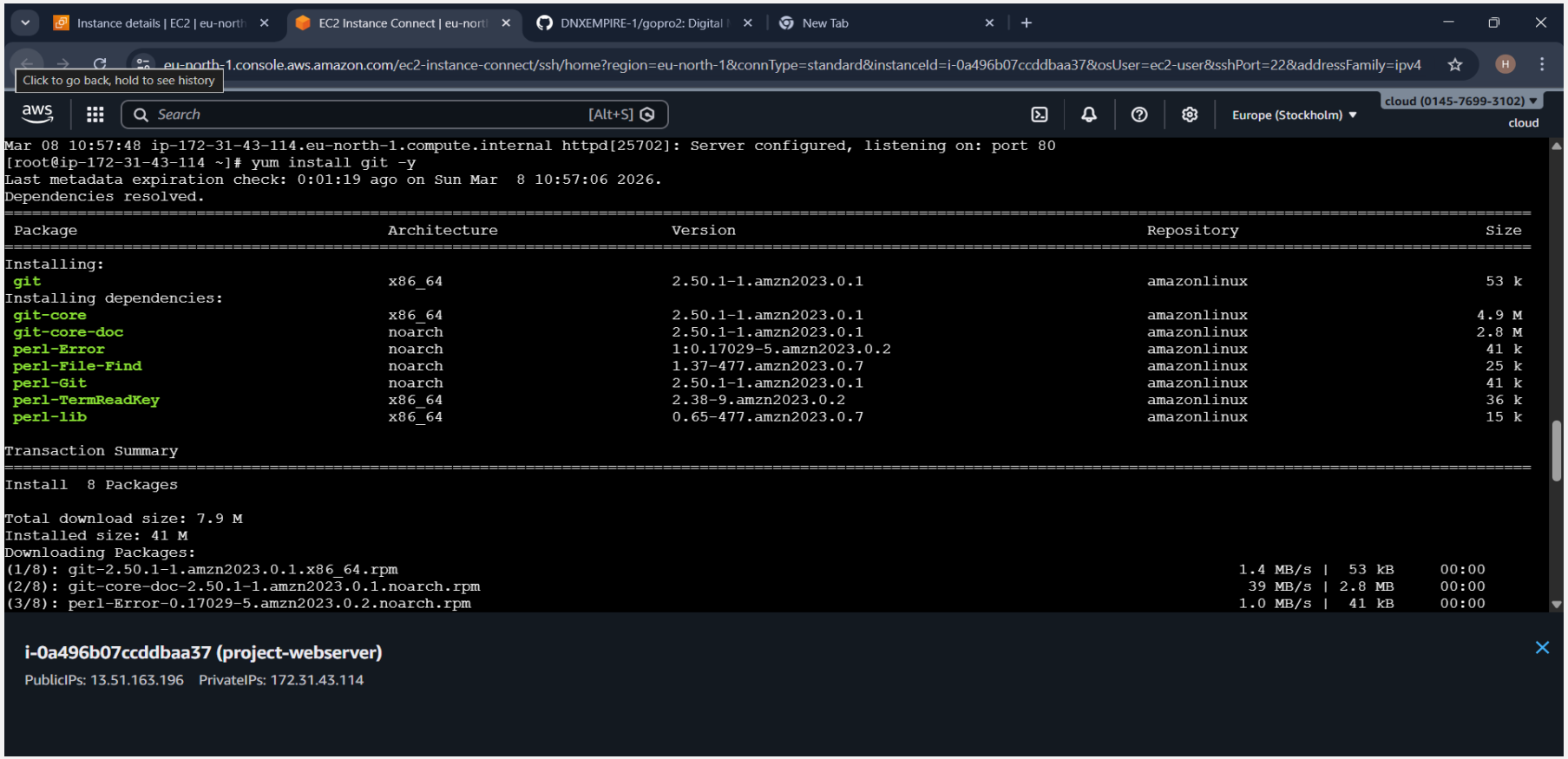
➢ We create EC2 instance.



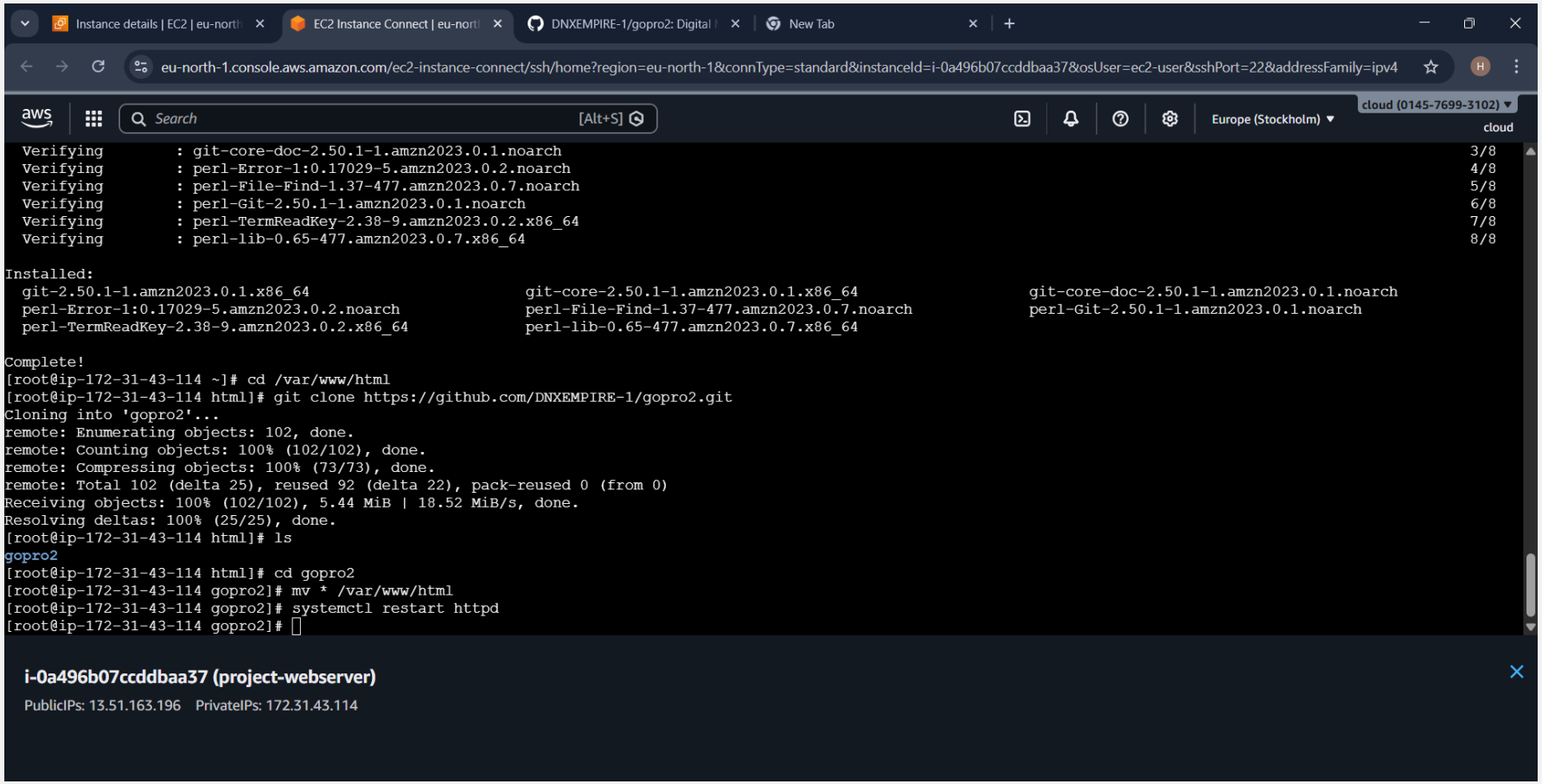
➢ Connect to instance using ssh and install httpd



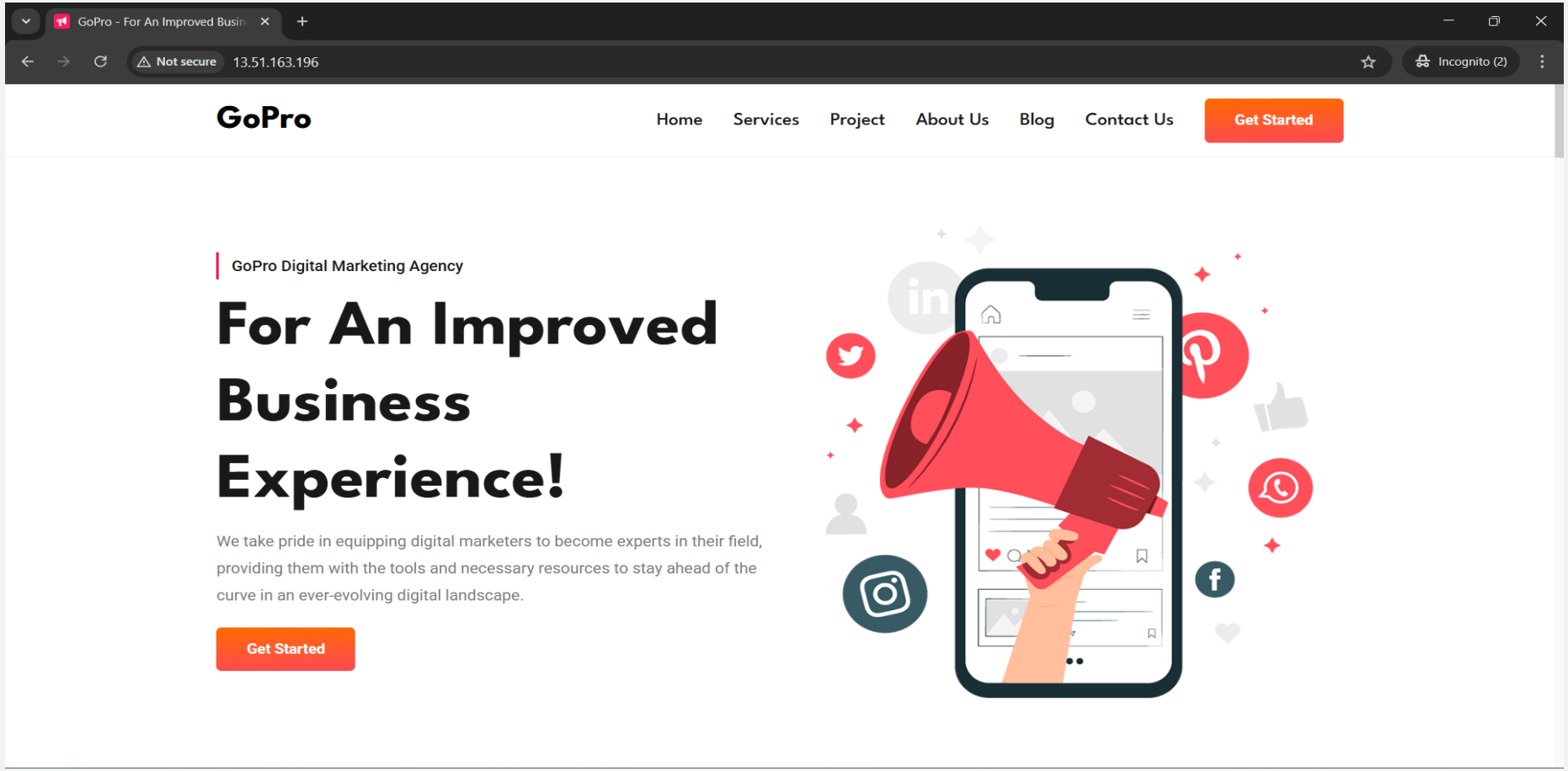
➢ We install git.



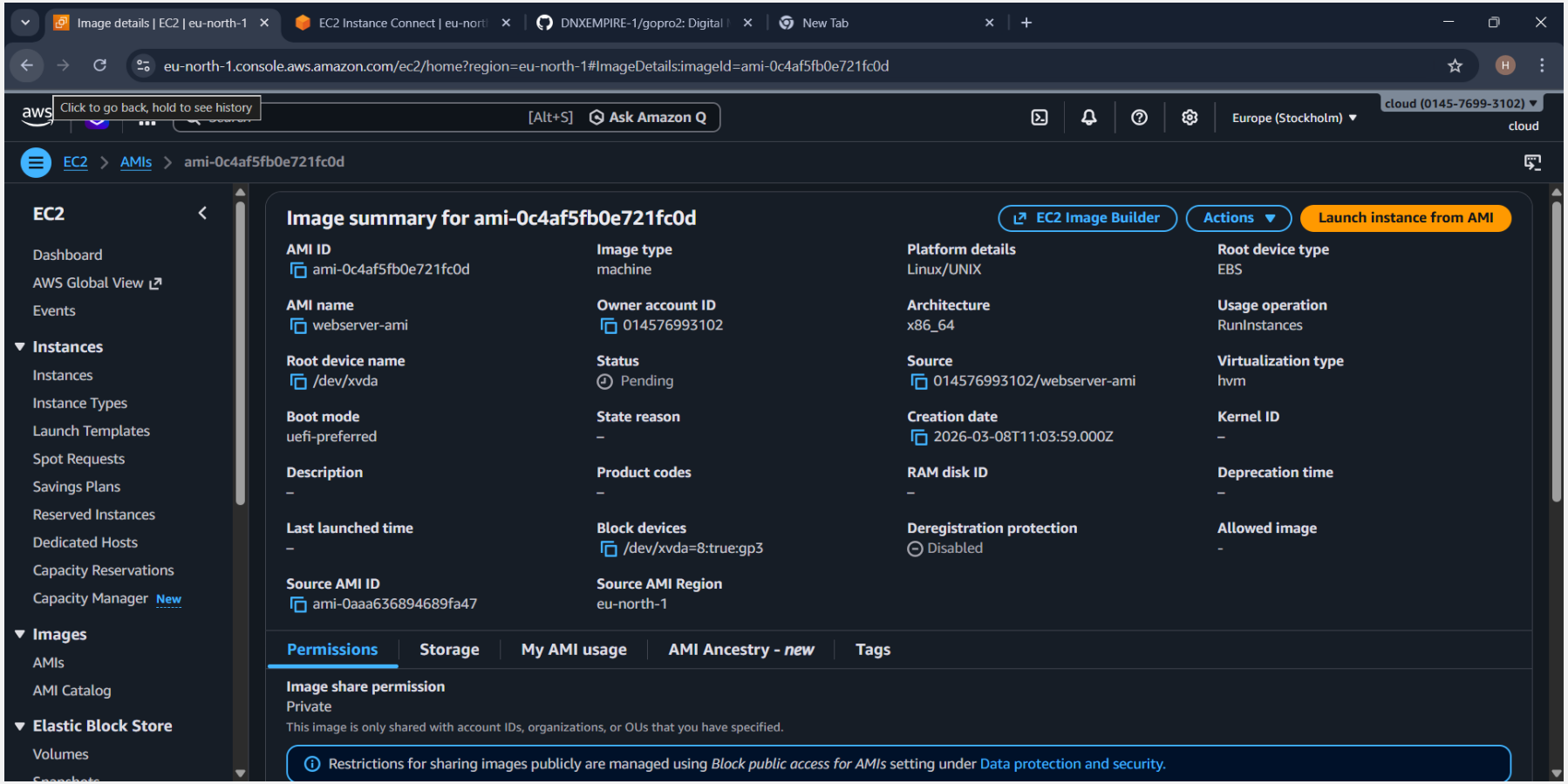
➢ We copy git URL and we use git clone in instance and move into /var/www/html directory restart httpd.



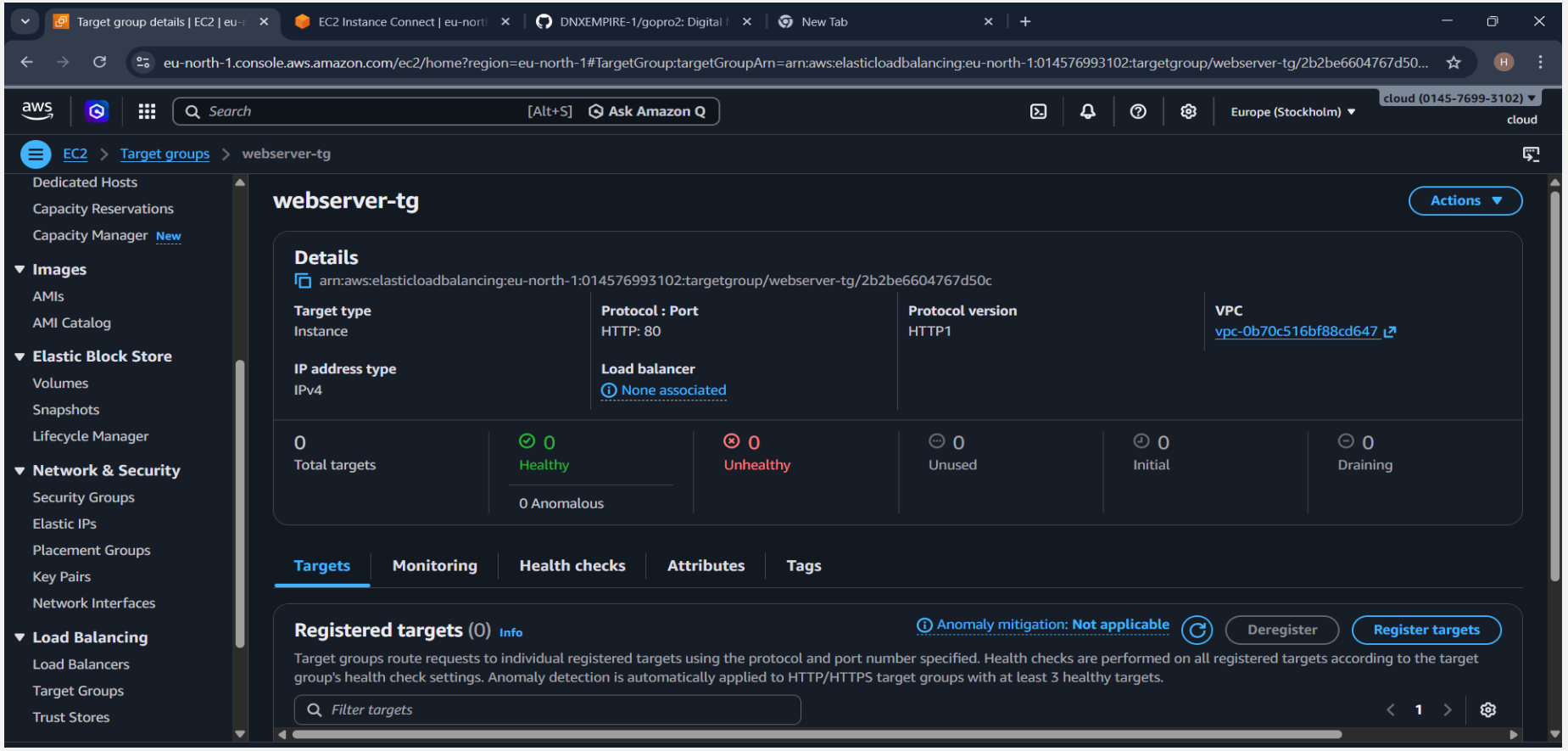
➢ By using public Ip address we can access our web application.



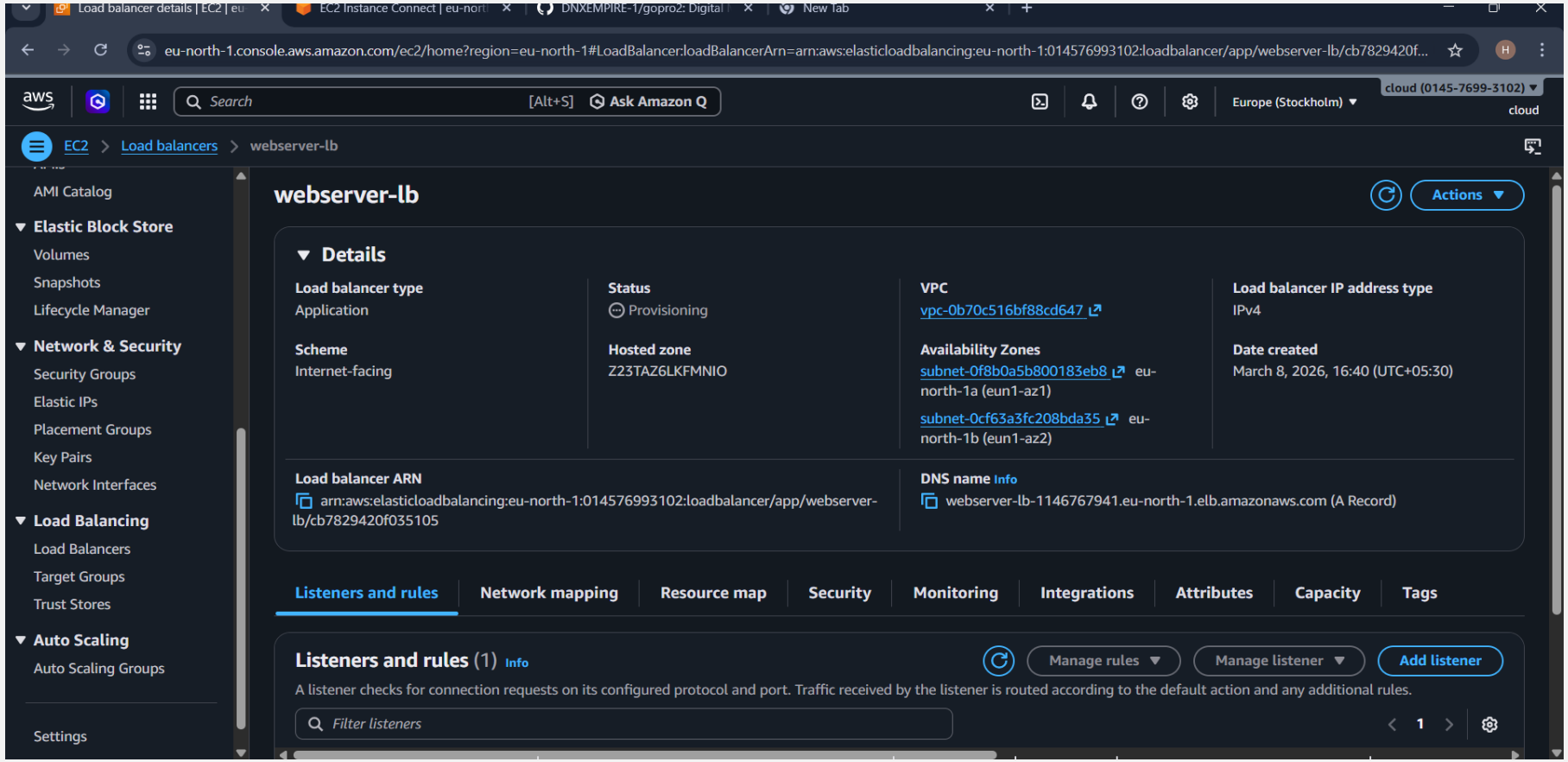
➢ Create ami(amazon machine image) with the help of EC2 instance(project-webserver).



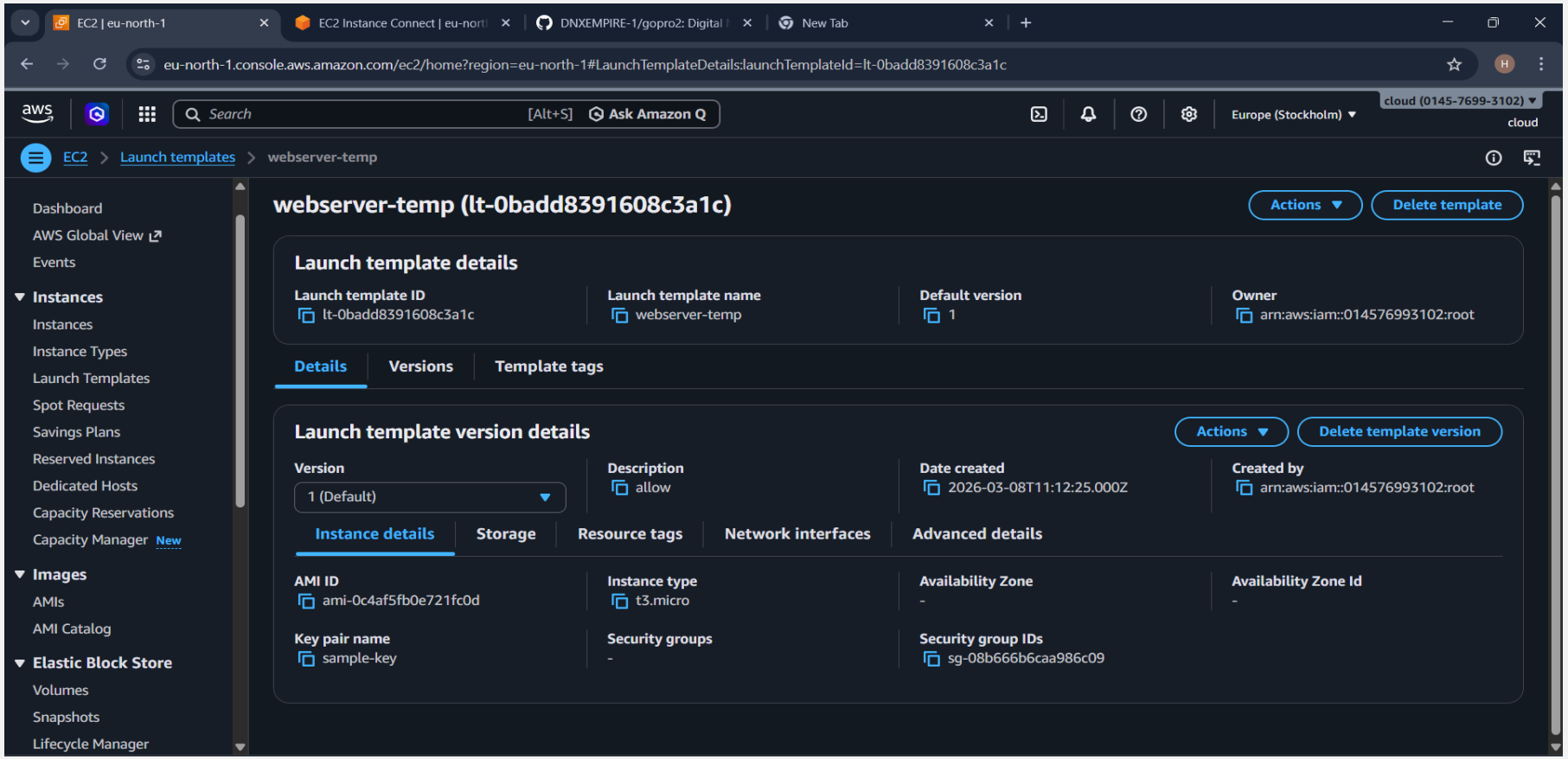
➢ Create target group we allow http 80



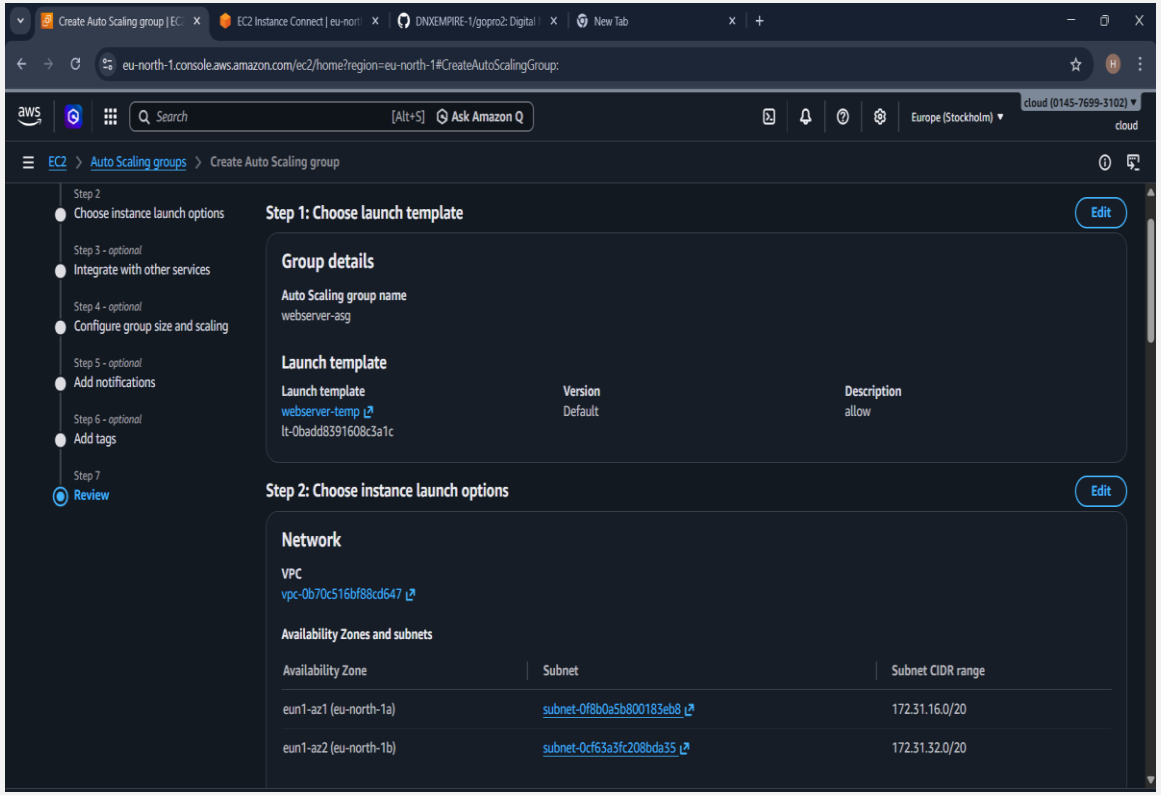
➢ Create application load balancer and attach target group.

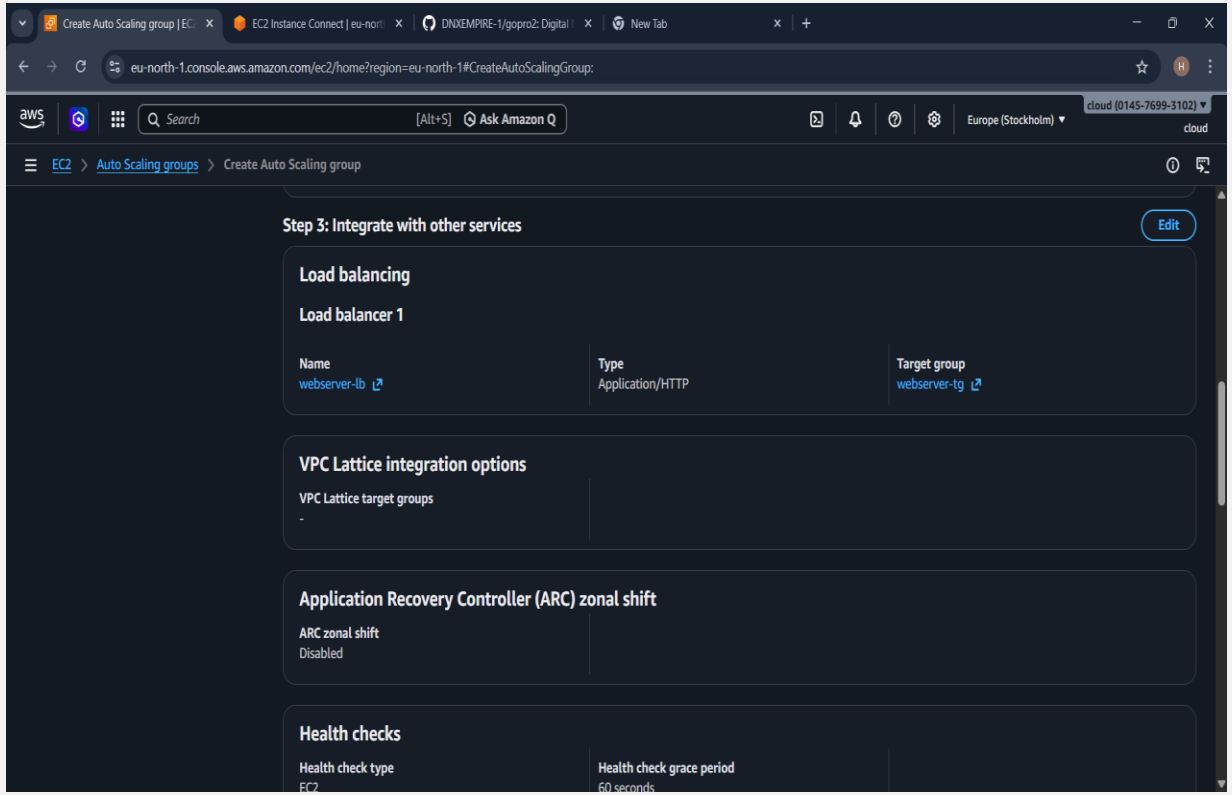


➢ Create launch template by selecting ami you created.

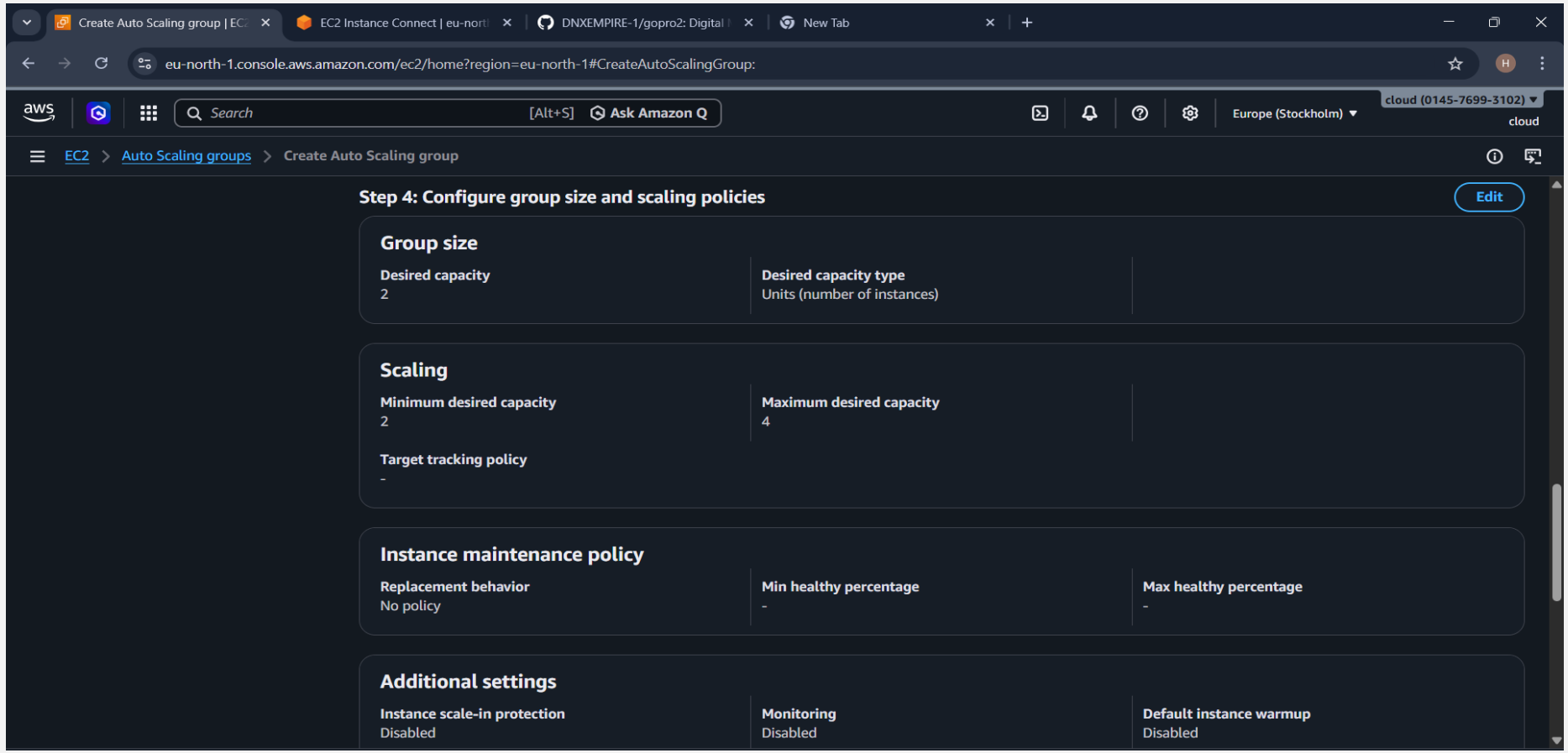


➢ CREATE AUTO SCALING GROUP.

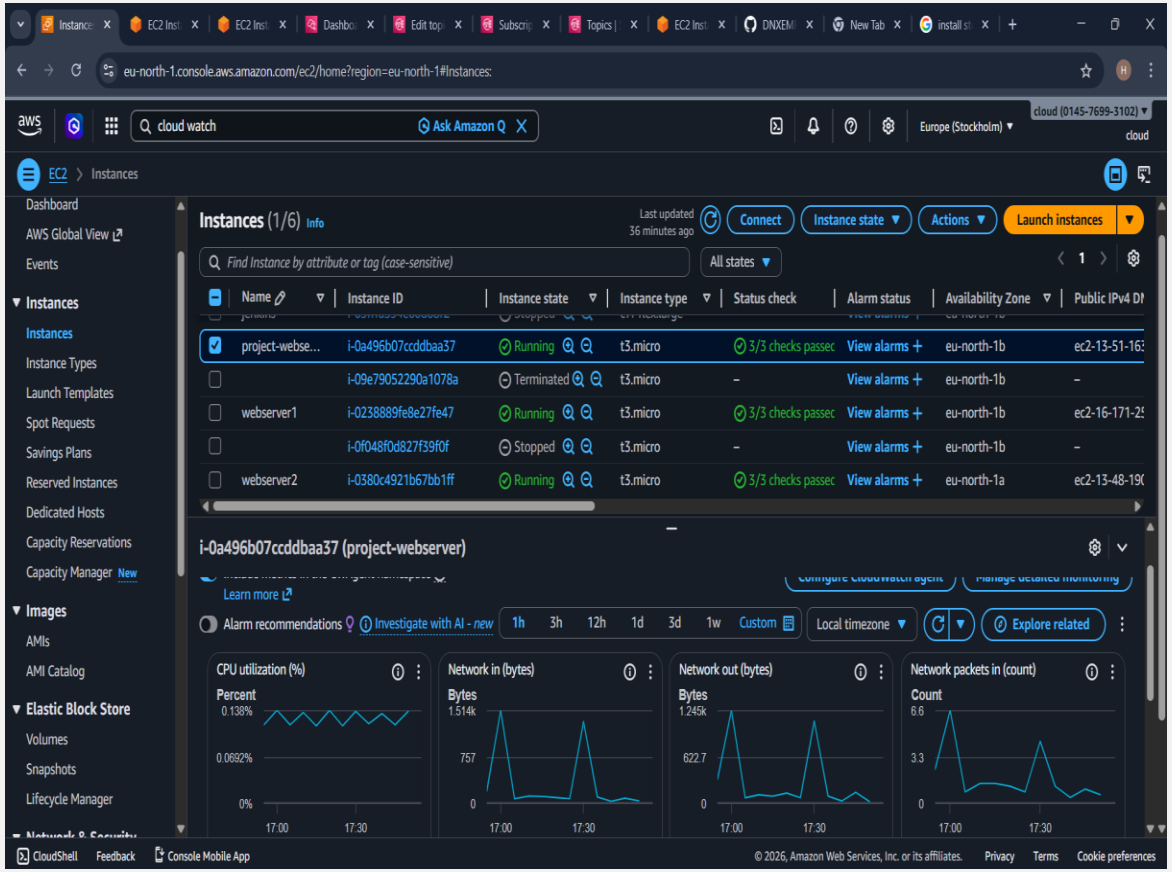


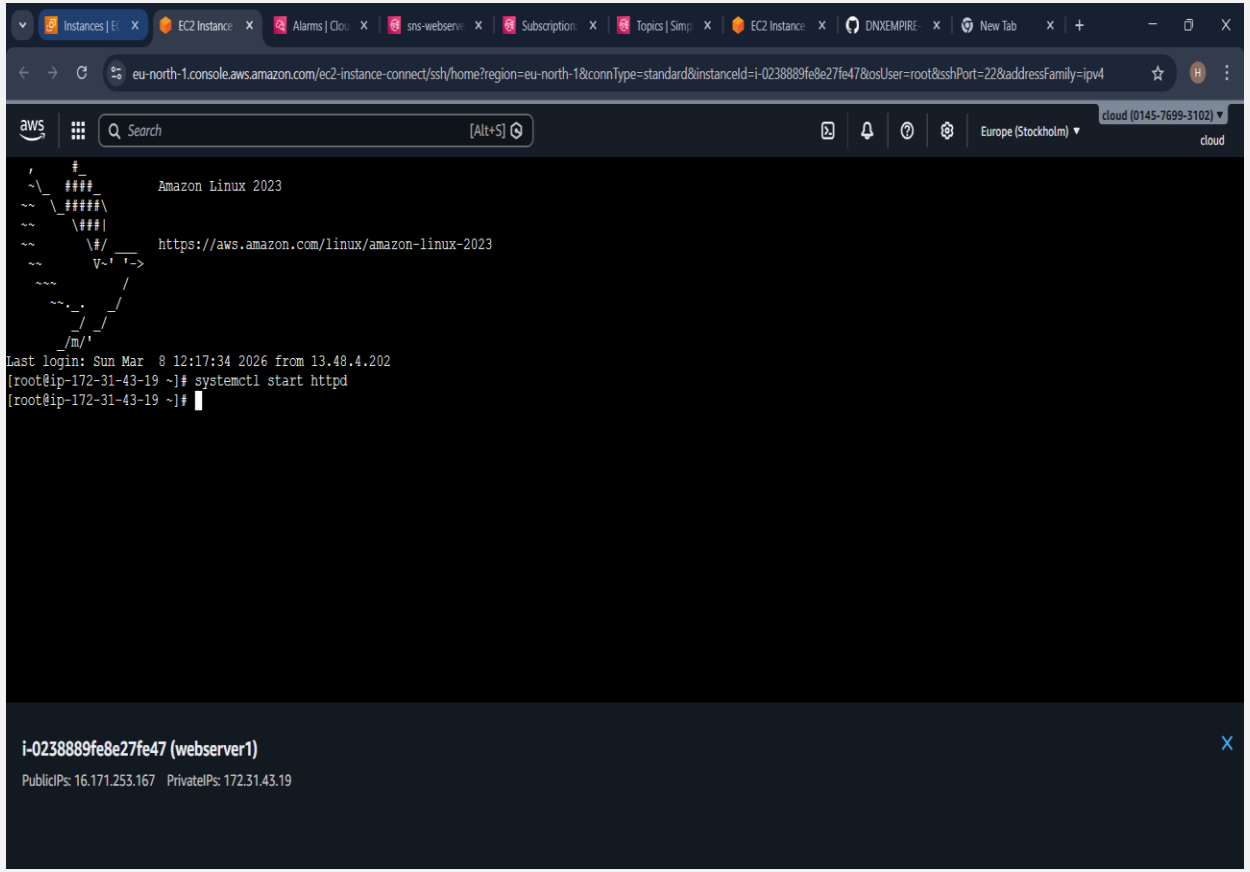


➢ CONFIGURE SCALING SETTINGS.

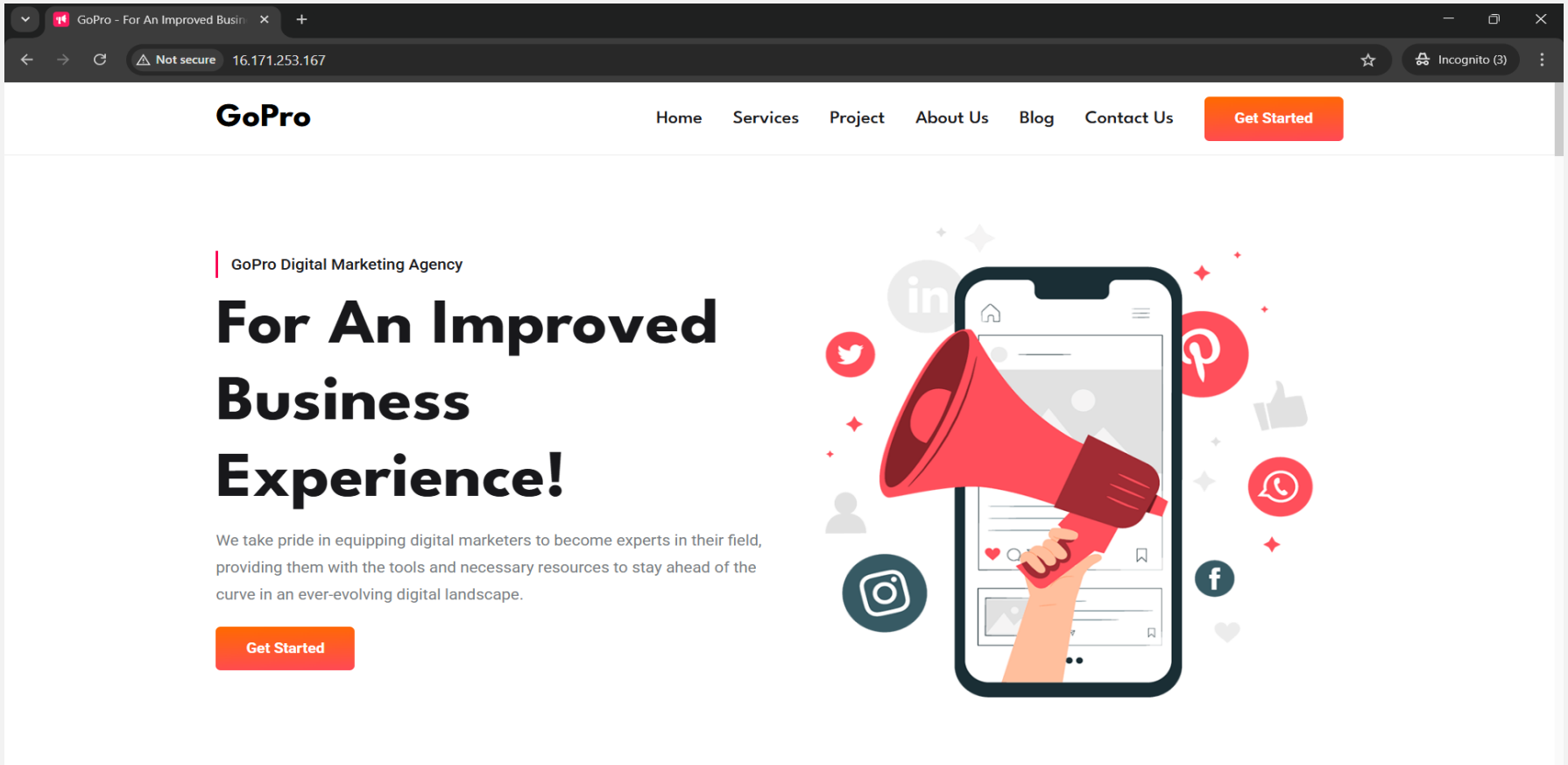


➢ After we create load balancer and auto scale instance will automatically implementand we check its working or not.

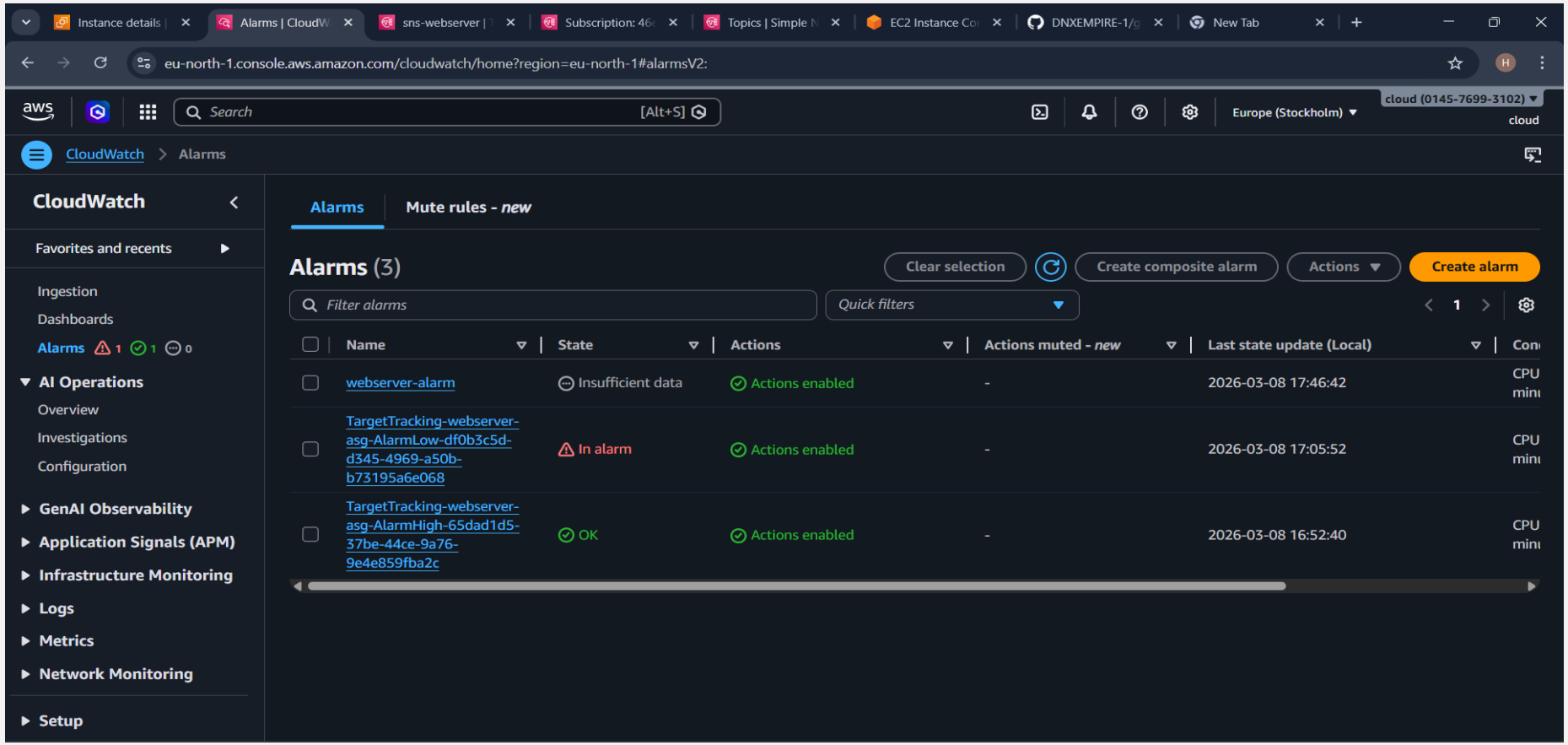




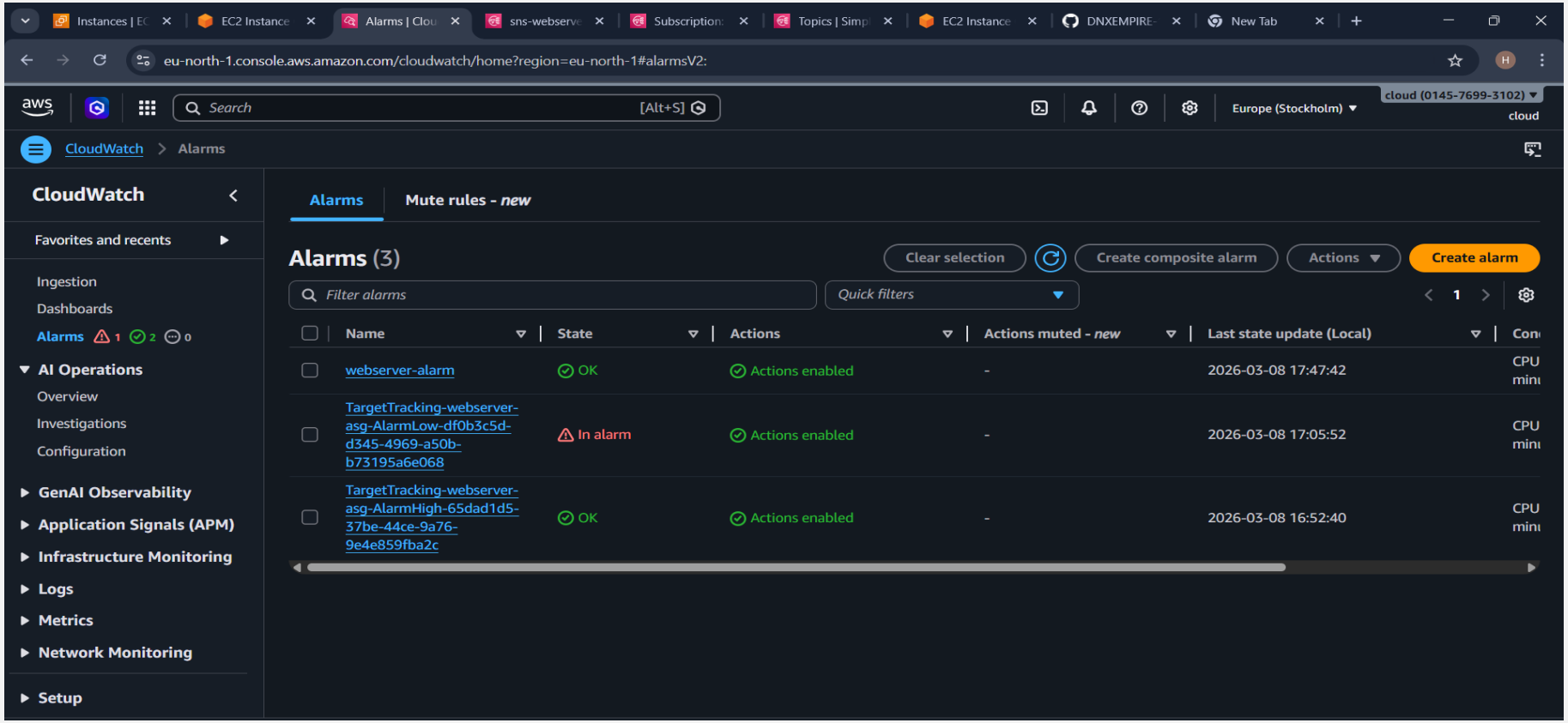
➢ We access on browser with the help of public ip.



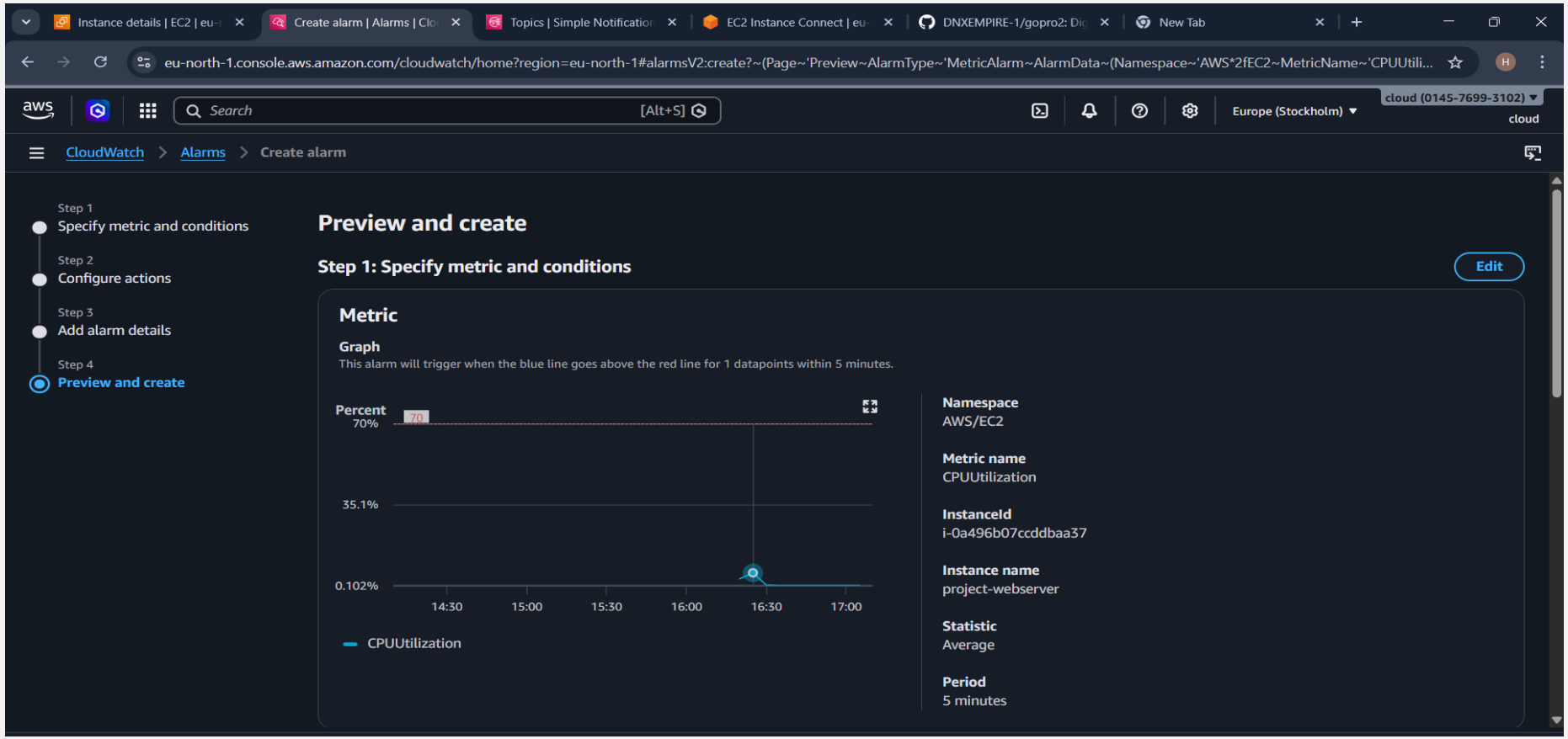
- Create cloudwatch alarm.
  Here insufficient data is alarm has just started metric is not available



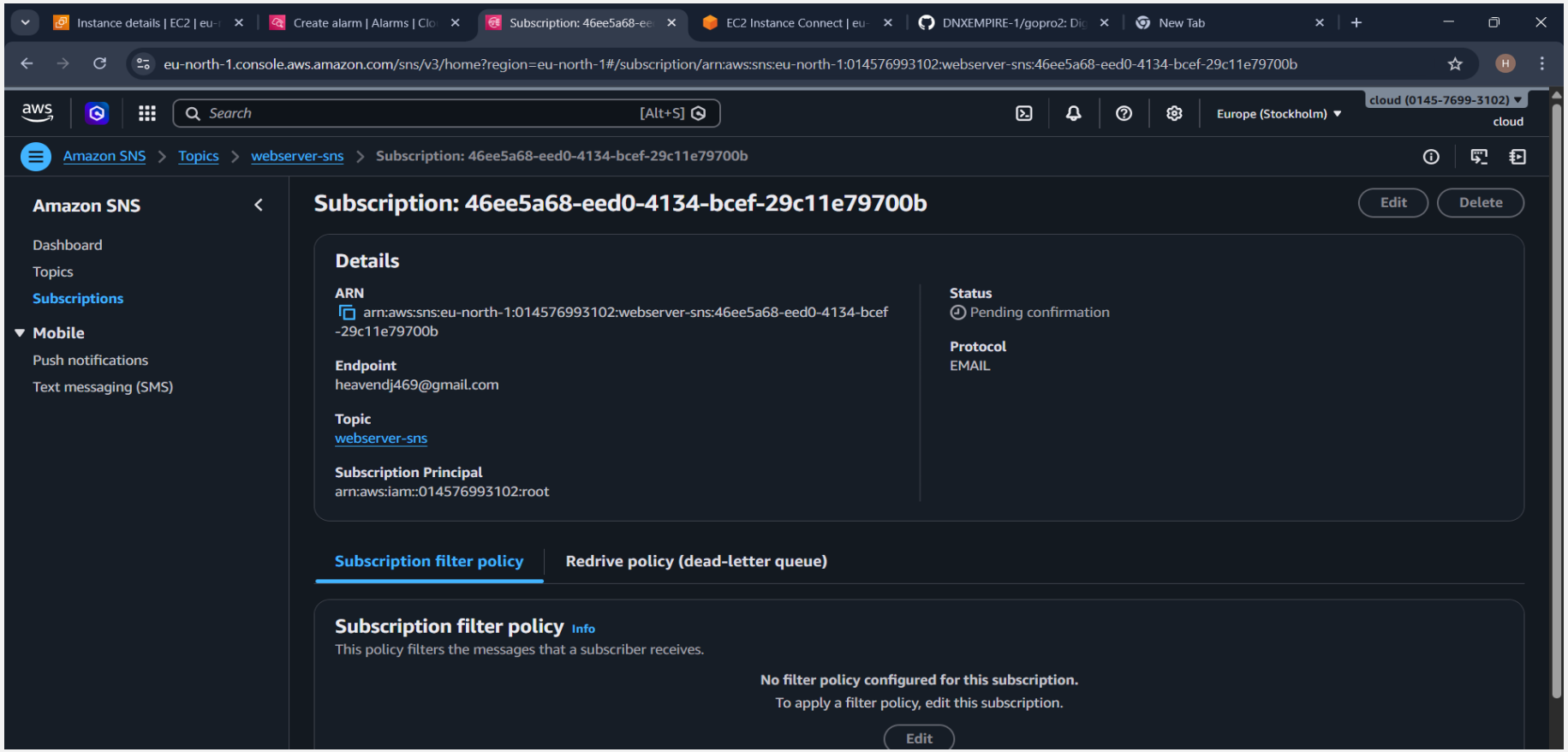
- Then it shows ok means the metric is with in the defined threshold.



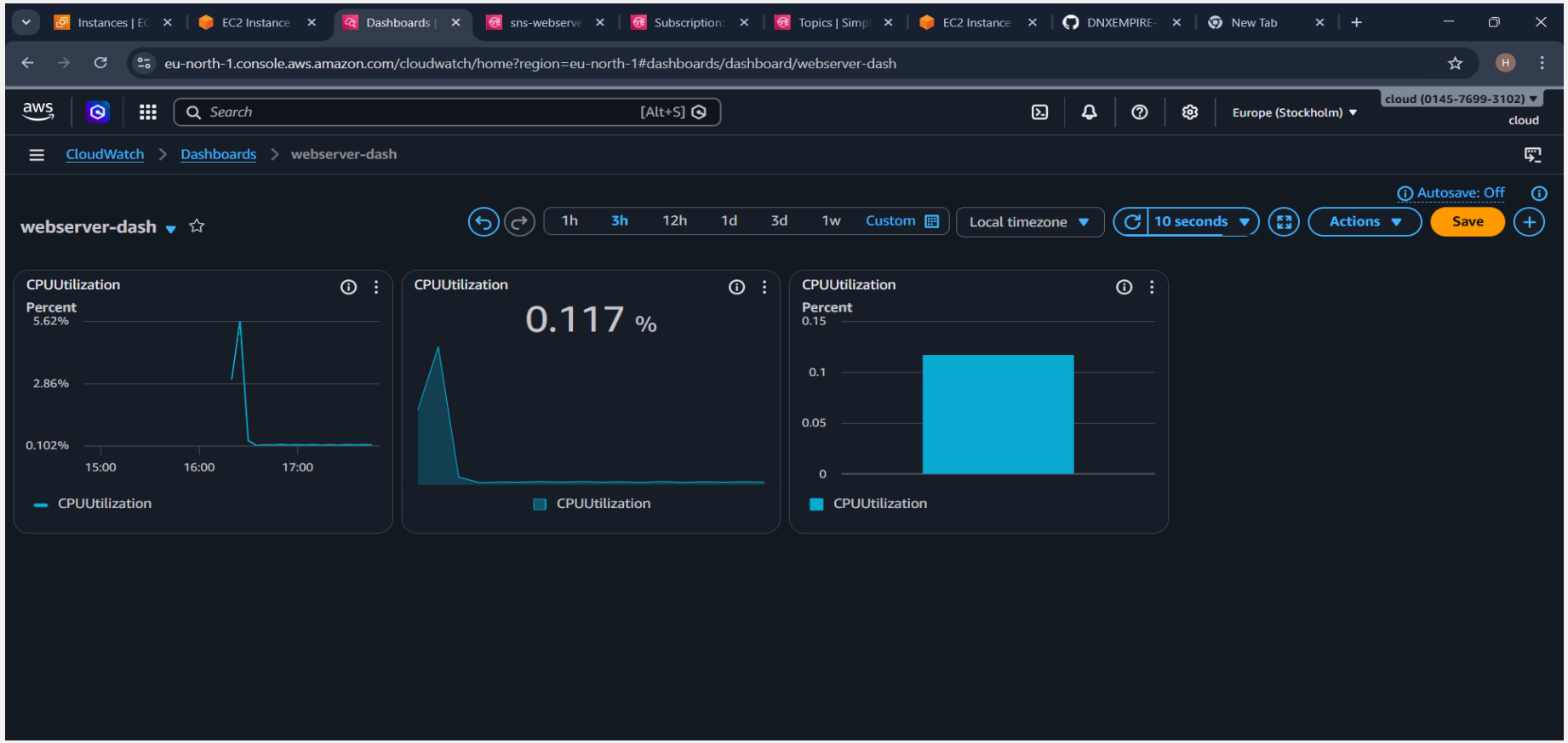
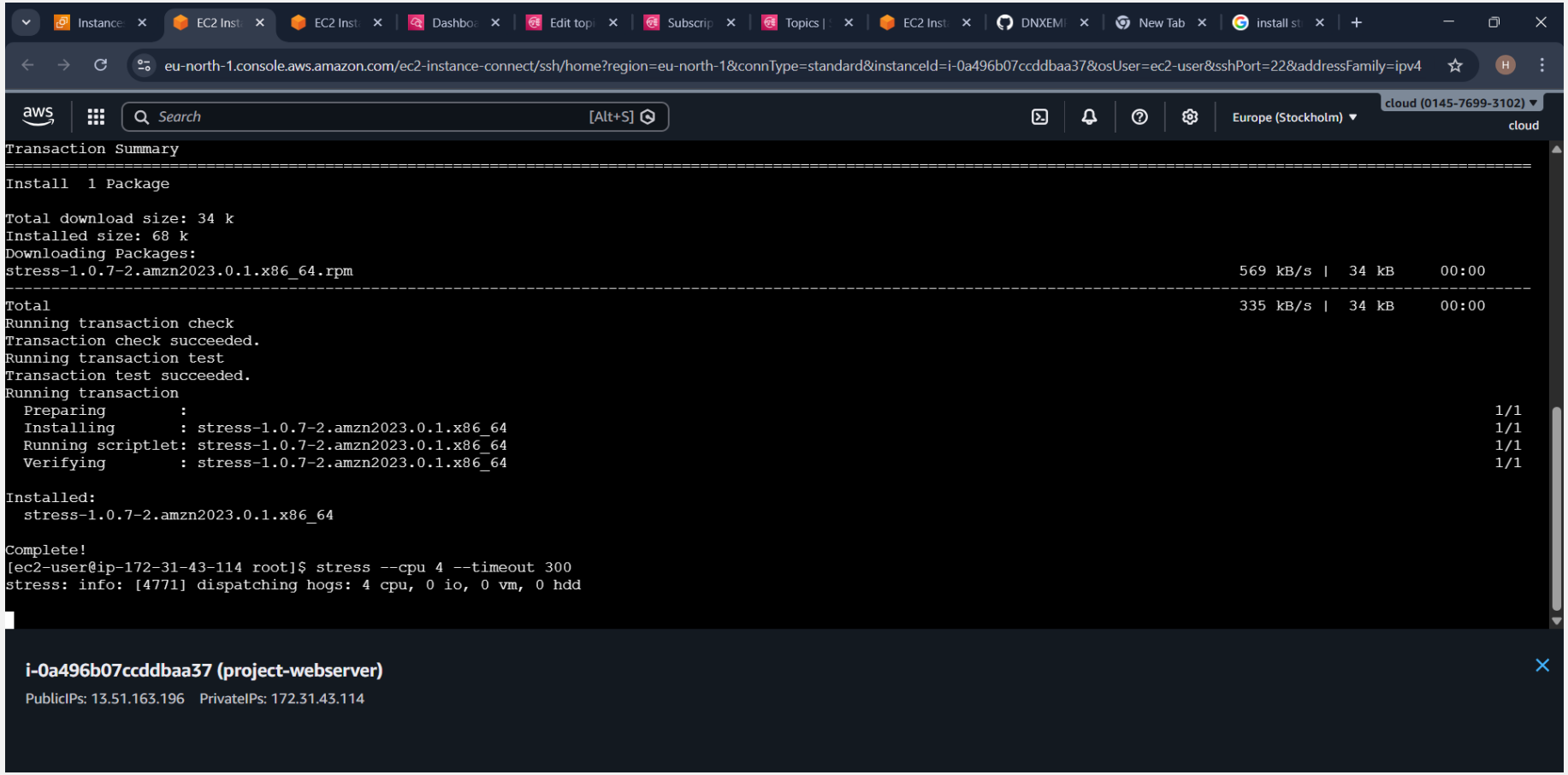
➢ Create sns topic(alarm system).



➢ WE MADE SUBCRIPTOIN.



- We create dashboard also it will show how cpu utilization.



➢ We put stress to instance but it will divided stress to multiple instance.